Advanced communications

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Introduction

In this exercise we are suppose to implement the basic cellular concepts and to design a communication system which only support GSM system. In this assignment it is mentioned to determine the best locations of the antenna's placement in the cell. The cell in wireless communication system is a smaller part of the larger area. A large area in cellular communication system is divided into different smaller parts but with same area and same specific shape to maintain the uniformity in the communication system and to avoid interference. One major question in this task is how to decide the location of the antenna.it depends on various factors i-e propagation loss, fading, interference, traffic capacity and the environment which includes what types of buildings are installed like if there are more high rise buildings or its some kind of rural area. Generally the shape of the cell is like a hexagon but it is not taken as exactly like hexagon to simplify the estimations and calculations. although calculated values from this cell is very accurate.

To solve the problem of spectral congestion the cellular communication system being designed.to achieve the high capacity in wireless communication system the cell size of the larger area is restricted to smaller cell sizes.in wireless communication we replace the low power antennas with high power antennas. The total portion of total cells.in this system same kinds of frequencies are reused with in same general system of base stations. The interference level is kept below the acceptable level.to enhance the capacity of both channel usage and utilization of more number of users is considered. The major property of the cellular communication system that it serve as a more million subscribes with limited number of frequency channels. the area of the city is divided on the basis of frequency reuse coverage.to reuse the frequency in different tiers the range of the antennas are also kept limited.

Radio telecommunication is one of important term, albert Einstein define it a long wire or long cat. As its like a tunnel. If you speak in one end and due to echo you will hear sounds on another end. Similarly, if we send some kind of signal form one side of tunnel and it will go to the other side without going anywhere. Also, there no such cats so we need to design our own communication channel.

Main point of the wireless communication system is to link the wireless device and mobile devices together so the data can be shared among them and also to establish and accommodate many users in given same spectrum which is an efficient method of frequency reuse. In order to avoid delay and inter symbol interference the frequency re-use method is used in cell planning. Also, one important term is BASE STATION. The base station consists of antenna which is majorly responsible for the communication. Height of the antenna is very important factor, more the antenna more the transmission power that is more SINR, but add to much length will make the design costly.

There is another term call handoff which would be used when one user changes his position from one cell to another. The neighboring cells along with the main cells makes a thing called c lusters. Clusters are basically made for the reuse of frequency among channels. The shape of the cell could be of many types like it could be circle, it could be triangle , it could be a hexagon. If we talk on practical aspects the shape of the cell is not exactly like hexagon it is some irregular shape but kinds of a hexagon.in this type of cell share the antenna could be on the edge of the cells. The reuse factor of the frequency in wireless communication system is given by 1/N.in the larger area system when cells are divided in smaller cells

each cell is allocated with 1/N channels to operate. moreover it is appoint to be noted that the cell size in this procedure remains the same in overall process. How the neighbor cells are separated or by which factor. The angle used in hexagonal geometry is separated by 600. The N of the cluster can only have the equation N=i2+ij+j2. Major the sizes of the cluster are preferred to be 12,7,5,4/In gsm normally the channel are taken in number as N=7.AND I=3,J=2 AND N =19.

Task 2

PLANNING wireless communication system practically lays off many trade offs.one of the major problem is the hand off between the cells. If the mobile user is moving with fast speed or high velocity t

Practical Handoff Considerations ν Problems occur because of a large range of mobile velocities θ ν ν ν pedestrian vs. vehicle user Small cell sizes and/or micro-cells → larger # handoffs MSC load is heavy when high speed users are passed between very small cells Umbrella Cells θ θ θ θ θ use different antenna heights and Tx power levels to provide large and small cell coverage multiple antennas & Tx can be co-located at single location if necessary (saves on obtaining new tower licenses) large cell → high

speed traffic → fewer handoffs small cell → low speed traffic example areas: interstate highway passing through urban centre, office park, or nearby shopping mall.

To cater with the high speed moving mobile users the umbrella directional antennas are used so they can cover the larger area in some specific direction.

Plotting the loss exponent v

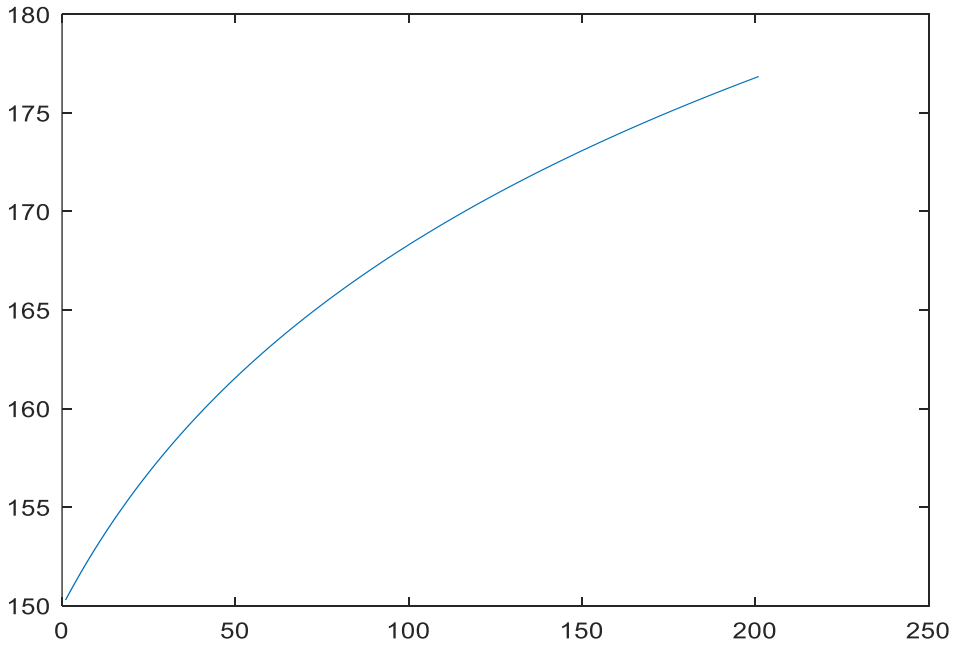
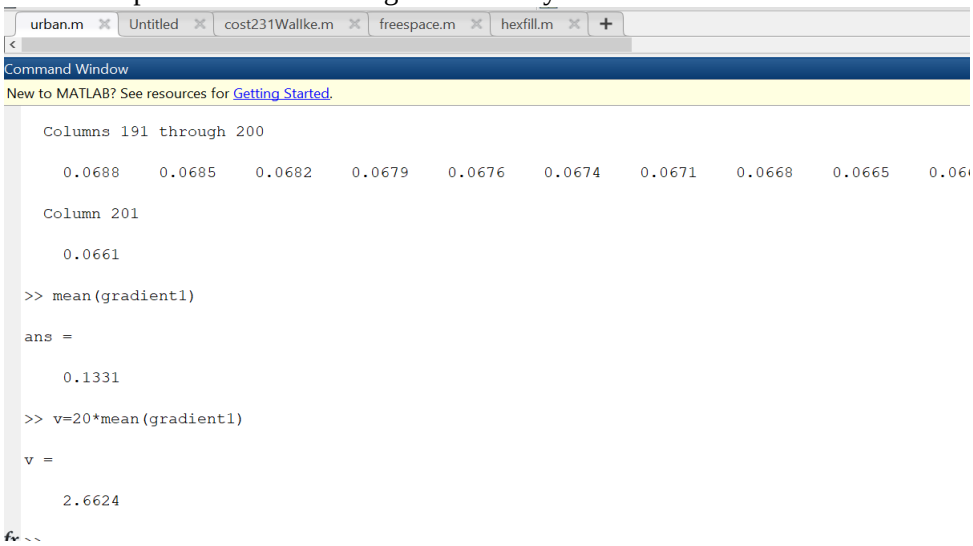


Figure 1

Path loss exponent v=2.664 taking v=n=2 in my case.



The value of 2.664 could be taken as 2 according to the following table of path loss exponent

Environment	Path Loss Exponent
Free space	2
Urban area cellular radio	2.7 to 3.5
Shadowed urban cellular radio	3 to 5
In building line-of-site	1.6 to 1.8
Obstructed in building	4 to 6
Obstructed in factories	2 to 3

Figure 2

Sigma =2.34

Taking N=5 and N=7 as two values can be taken to evaluate the network.

Calculating Signal To interference Ratio.:

$$\frac{S}{I} = \frac{(\sqrt{3N})^n}{6 \cdot \sum_{k=0}^{K} k^{1-n}}$$

– Considering *K* layers of interfering cells

– For *N* fixed, *n=2* and the number of layers K→∞; S/I →0

$$I = \lim_{K \to \infty} O(\sum_{k=0}^{K} \frac{1}{k}) = \infty$$

N=5,n=2, layers of interfering cells = 2 as number of channel can be from 0-6.

SIR=$(\sqrt{3 * 5})^2/6(0^{-1} + 1^{-1} + 2^{-1})$=280 dB

The number of channels can be found by using grade of service formula. As c is the number of the channels in a trunked syste, where Ac=U*Au/C

The interference is of two types majorly. First is co channel interference and the second is adjacent channel interference also known as acci mostly caused by the wrong useage of frequency reuse factor.

Signal to interference ration S/I in which S is the desired signal power where I is the interference.i could also be labelled as the interference power from the ith co channel cell. The average power received at some sepcific distance d will Pr=Po(d/do)^n.The tier terminalogy is introduced to reuse the freqeuency

Example are taken from online platform to discuss the SIR.

S/I includse the cluster size N=4 (choices 4,7,12) path loss exponent (means) n=4 co-channel reuse ratio Q= sqrt(3N)=3.464102 Ratio of distance to radius Q=D/R=3.464102 number of neighboring cells io=6 ,# of sides of hexagon signal to interference ratio S/I= (D/R)^n / io =24 convert to dB, S/I= 10log(S/I)=13.80211dB S/I is less than required, it will not work! cluster size N=7 path loss exponent n=3 Q=sqrt(3N)=4.582576 number of neighboring cells io= 6, # of sides of hexagon signal to interference ratio=S/I= (D/R)^n / io =16.03901 convert to dB, S/I=10log(S/I)=12.05178dB S/I is less than required, it will not work! 39

Adjacent Channel Interference includes the Results from imperfect receiver filters, allowing nearby frequencies to leak into pass-band. Can be minimized by careful filtering and channel assignments. Channels are assigned such that frequency separations between channels are maximized. For example, by sequentially assigning adjacent bands to different cells Total 832 channels, divided into two groups with 416 channels each. Out of 416, 395 are voice and 21 are control channels. 395 channels are divided into 21 subsets, each containing almost 19 channels, with closet channel 21 channels away If N=7 is used, each cell uses 3 subsets, assigned in such a way that each channel in a cell is 7 channels away.

Grade of Service:

The concept of trunking has been introduced to tackle with the lmitation of channel usage.the unit of the traffic is called Erlang.in the grade of service there are two types of trunked system.in which first one can be name as trunked system when the calls is blocked and the other is when the call is delayed.

In trunking system the million number of users are allowed to utilize very few gateways or the channel efficiently. The system of call termination works on the grade of service.the user in the gsm system basically subscribe the service for ther usage property as the user thinks that whenever he wants to make a call he will be offered a channel.if he is nt offered a channel then the service would be termed as bad service by the network and this problem persists the user might change the network or the services offered by the telecom operator.there are many entities that .re used in the grade of service in which one majore term is setup time. Setup time is the time taken to maange the call.or to allocate the freqeuncy channel now how the trunking is divided into two categoreis such that fist the call is blocked.when the user makes the call and its is not reponseive by the system or the system does not allocate a channel to the user due to unavaialbilty of the channel the call is blocked this can be termed as congestion. Or also called holding time in which the call is being blocked by the MSC.THE EVERAGE number of called request per time is called a lamda λ.THE traffic intensity is measured in the unit called erlangs.

The user when does not given the channel then we can say that calls are blocked now on the contrary the second term is defined as when the users are given the channel but with some delay this term is called block calls are delayed. There is one draw back in this system that when the user makes the call the system makes the user to wait for some time .as this time is taken by the msc to make the switching of channel .as availably of channel could be in the neighboring cells. the time taken for the holding the call of the user is called dwell time. Moreover there is a point to be noted in this section that the blocked calls even after the dealay gets blocked again this is the most severe condition offered by the channels and user is allowed to try again. The erlang formula is based on the assumption that all channels are fully available to be called on anytime in the system.infact in practical the channels are not free they might be occupied by some Erlang c formula most probably gives the probability of a call not having the immediate access to all the channels

To explain this term the example is taken from an online platform. For example, if a group of 100 users made 30 calls in one hour, and each call had an average call duration(holding time) of 5 minutes, then the number of Erlangs this represents is worked out as follows: ν Minutes of traffic in the hour = number of calls x duration Minutes of traffic in the hour = 30 x 5 = 150 Hours of traffic in the hour = 150 / 60 = 2.5 Traffic Intensity= 2.5 Erlangs.

Traffic concepts in trunking explained by different terms . the total traffic intensity per cuser is denoted by Au, and Au= λH. Where h is the number of duration of the call in seconds.

Task 3

Starting the urban.m file and running the city vm187

```
>> urban
Please specify the input filename for building data (without .bldg extension ):
city_vm187

ans =

251 buildings read in.

Please click the 'Add' button to add antennas.
Then click the 'Save' button to write antenna data to file.
The 'Calc' button runs the propagation model.
```

Estimating the power loss exponent V by using function cost231WalIke.m

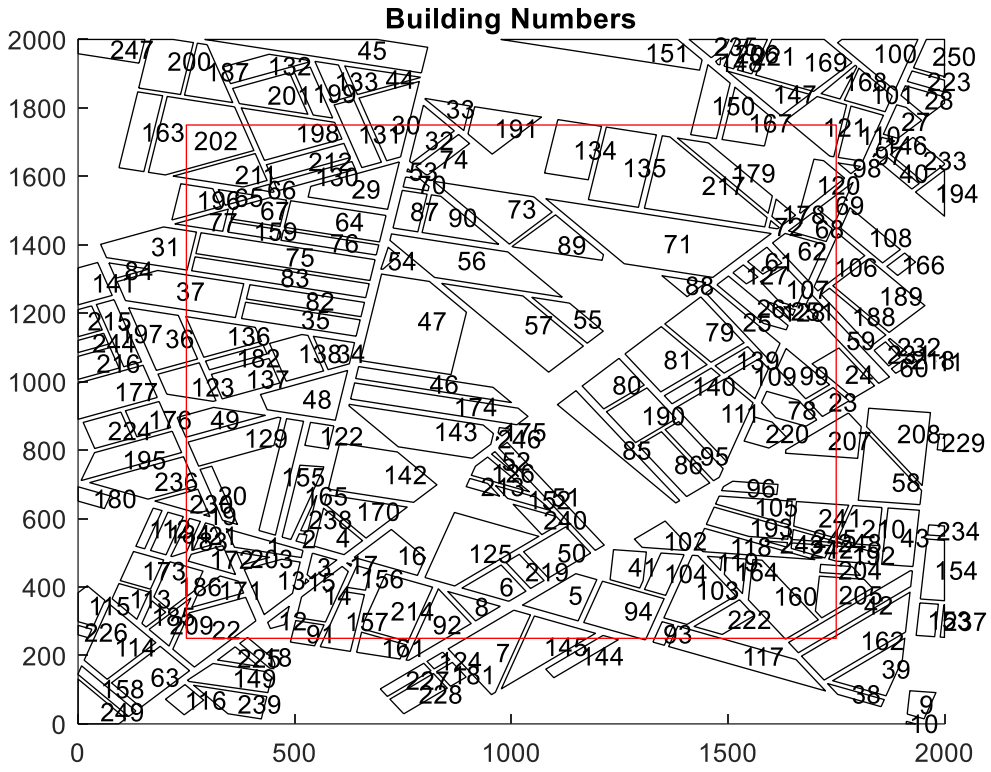


Figure 3

Placing the base stations on different sites

Place of antennas

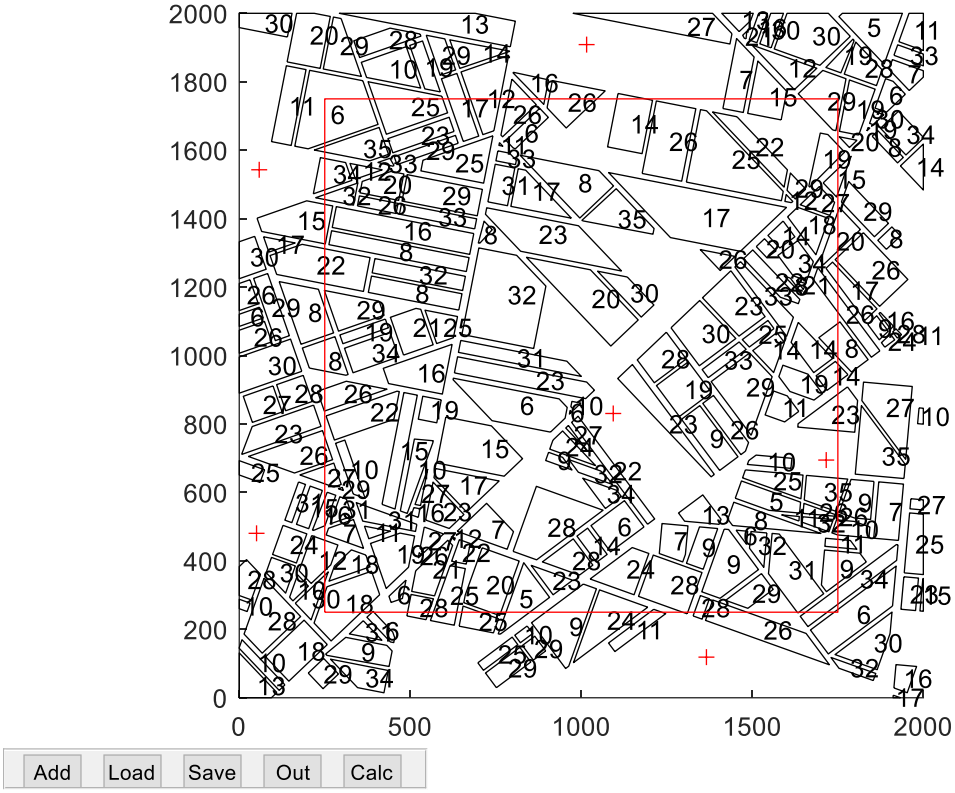


Figure 4

Antennas adjusted by using hellfix.m and more antennas are automatically adjusted

Using all the constraints and calculated values in task 2

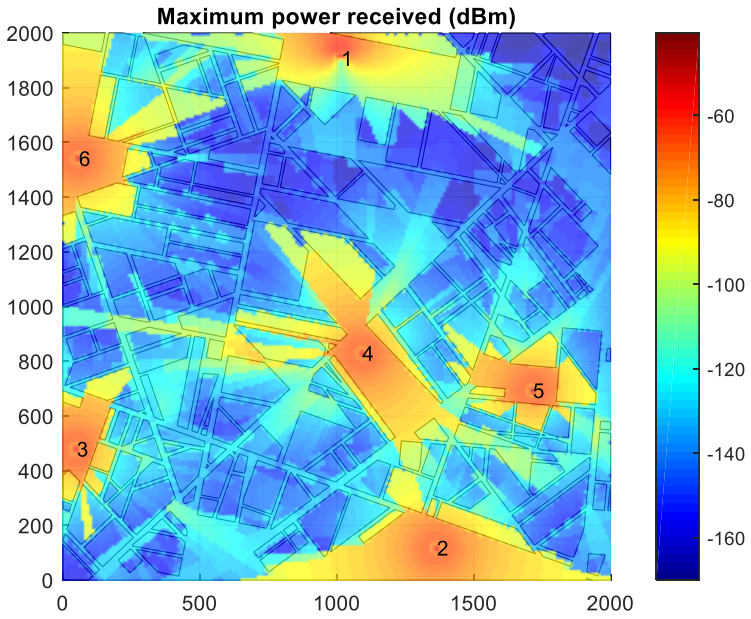


Figure 5

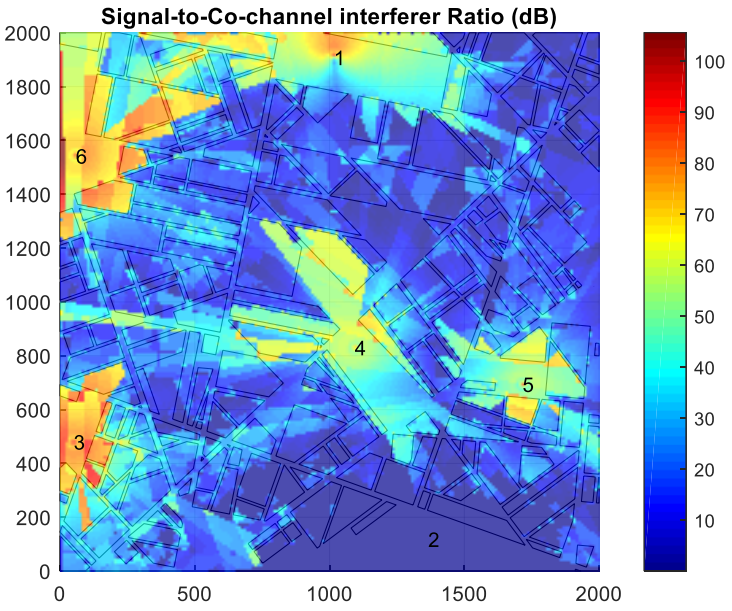


Figure 6

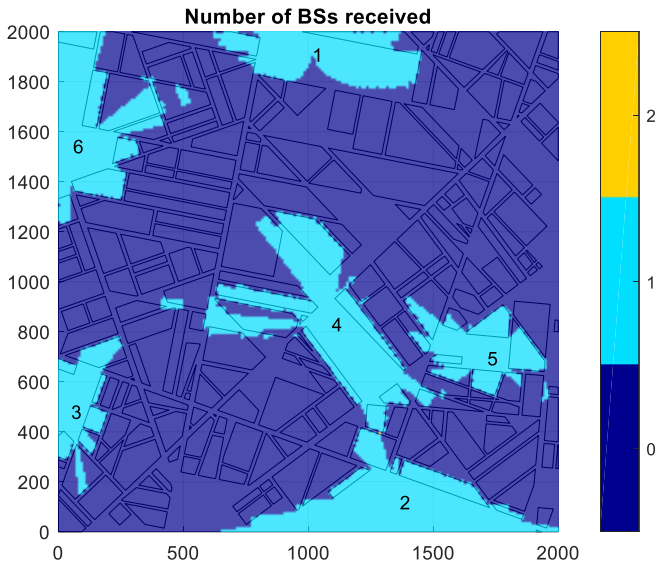


Figure 7

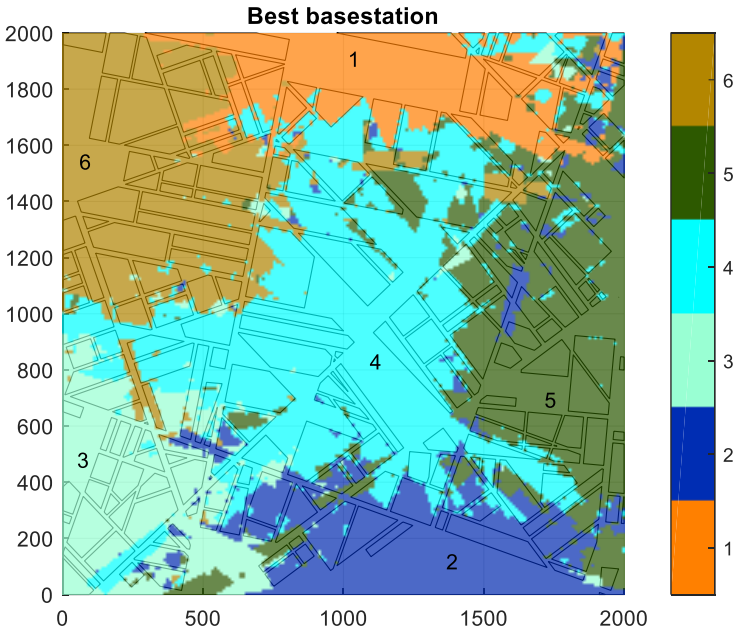


Figure 8

In the end we can say that all the theoretical values calculated are near to estimated values .the data taken from the map is not fixed as with the change in the antenna property or parameters the outputs on the graphs are changed